



Instructor: Date:

Learner: Regional Branch:

Program: Center:

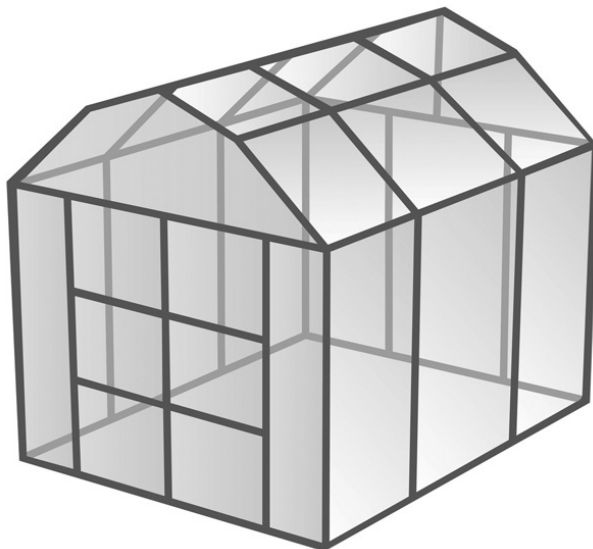
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Let's Start!

What do a greenhouse and the earth have in common?



Learning Outcome

Describe environmental change in social contexts.

Ask your partner:

- Are greenhouses common in your area? Why or why not?
- How do you think the science of a greenhouse applies to the earth's atmosphere?

Can Do Statements

- I can describe social and urban change, and environmental problems.
- I can talk about cause and effect, and hypothetical situations.
- I can talk about things that are happening and things that have happened.





Let's Explore!

The atmosphere of the earth is a comfortable 59 degrees **Fahrenheit***.

* "Fahrenheit" is the temperature scale that is used in the US $59^{\circ}\text{F} = 15^{\circ}\text{C}$.



Instructions: Read the text aloud.

If you have ever tried to grow plants, you know that plants need a warm environment. For this reason, plants are being grown in glass greenhouses. The glass traps the warm air inside so that the plants can grow.

This happens on the earth as well. Gases found naturally in the atmosphere trap the heat of the sun, and warm up the earth to a comfortable 59 degrees Fahrenheit. This has been called the "greenhouse effect", which makes it possible for humans, animals, and plants to live. If there were no gases in the atmosphere, the earth would be a frozen snowball.

Many scientists say that human activities have caused greenhouse gases, such as carbon dioxide, to build up in the atmosphere. According to this theory, whenever we burn fossil fuels, such as gasoline, oil, coal, or natural gas, more greenhouse gases are being released into the atmosphere. As a result, the atmosphere is trapping more heat, and the earth is warming up. This theory is known as global warming.

Some scientists claim that global warming will take generations to affect the earth. Others predict that if we do not reduce the release of greenhouse gases now, global temperatures will rise several degrees within the next 100 years. This change in temperature could mean major climate changes throughout the world.

Scientists claim that if we had acted earlier, humankind wouldn't have found themselves in this predicament. Once greenhouse gases have been released into the atmosphere, most of them remain there for decades and continue to trap heat.



Let's Practice!

What was done?

Instructions: Classify the sentences into the right category. Practice saying the sentences aloud. Check your pronunciation with your instructor.

- If you ever tried to grow plants, you know that ...
- Greenhouse gases are being released into the atmosphere.
- Scientists say human activities have caused greenhouse gases ...
- Once greenhouse gases have been released ...
- Plants are being grown in glass greenhouses.
- This has been called the "greenhouse effect".
- As a result, the atmosphere is trapping more heat.

Passive sentences

Active sentences



Let's Practice!

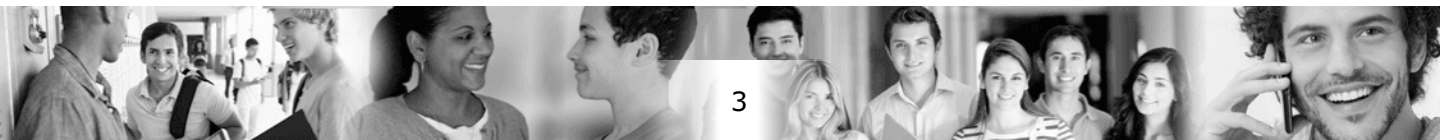
What would the earth be like?

Instructions: Complete the text below by choosing from the options you are given.

Word Bank

if we had acted earlier / have ever tried / were no gases / would be a /
wouldn't have found / you know that / temperatures would rise

- If there _____ in the atmosphere, the Earth _____ frozen snowball.
- Others had predicted that if we did not reduce the release of greenhouse gases now _____ several degrees within the next 100 years.
- Scientists claim that _____, humankind _____ themselves in this predicament.
- If you _____ to grow plants, _____ plants need a warm environment.





Extension Activity

Practice using the third conditional!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- On 8 small pieces of paper, write down 4 nouns related to the environment and four verbs related to the environment.
- With a partner, take turns choosing one noun and one verb and making third conditional sentences with them. For example: garbage, dispose = If I had disposed of my garbage, my house would not have been dirty.
- When you have finished your words, think of new sentences and discuss the conditions that you created.

Self-Assessment

Yes No

I talk about the environment.

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I talk about unreal conditions in the past.

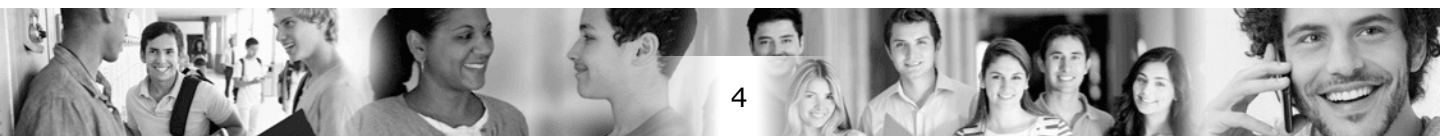
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I discuss the conditions with a partner.

☐ ☐


Target Words

- to grow** (verb): to increase in amount, size, number, or strength
e.g. Plants need a warm environment to grow in.
- greenhouse** (noun): a small glass building used to grow plants all year round
e.g. Plants are being grown in glass greenhouses.
- to trap** (verb): to prevent something from escaping from somewhere
e.g. The glass traps the warm air inside so that the plants can grow.
- snowball** (noun): a ball formed by packing snow tightly together
e.g. If there were no gases in the atmosphere, the Earth would be a frozen snowball.
- to build up** (verb): to construct
e.g. Human activities have caused carbon dioxide to build up in the atmosphere.
- fossil fuel** (noun): a combustible organic material, such as oil, coal, or natural gas
e.g. Whenever we burn fossil fuels, such as gasoline, oil, coal, or natural gas, more greenhouse gases are being released.
- coal** (noun): a hard black mineral burned to produce heat
e.g. Coal is a fossil fuel that causes global warming.
- release** (noun): allowing to flow out of a usual container
e.g. If we do not reduce the release of greenhouse gases now, global temperatures will rise several degrees within the next 100 years.
- predicament** (noun): a difficult or unpleasant situation where you have to make a difficult choice
e.g. If we had acted earlier, humankind wouldn't have found themselves in this predicament.





Learning Points

The Third Conditional

There are four types of conditional sentences in English:

1. Zero Conditional - to talk about certainty and facts
2. First Conditional - to talk about real possibilities in the present or the future
3. Second Conditional - to talk about *unreal* possibilities in the present or the future
4. Third Conditional - to talk about unreal *past* situations

You use third conditional to talk about unreal situations in the past. These are things that did not happen in the past. You often use the third conditional to talk about regrets.

Look at the chart below to compare the structure of each kind of conditional:

Conditional Name	Structure	Example
Zero	<i>If + Simple Present, Simple Present</i>	<i>If you heat ice, it melts.</i>
First	<i>If + Simple Present, will + Simple Present</i>	<i>If we do not reduce the release of greenhouse gases now, global temperatures will rise.</i>
Second	<i>If + Simple Past, would + Simple Present</i>	<i>If there were no gases in the atmosphere, the earth would be a frozen snowball.</i>
Third	<i>If + Past Perfect, would have + past participle</i>	<i>If we had acted earlier, humankind wouldn't have found themselves in this predicament.</i>

Examples:

- *If there were no gases in the atmosphere, the earth would be a frozen snowball.*
- *If we do not reduce the release of greenhouse gases now, global temperatures will rise several degrees within the next 100 years.*
- *Scientists claim that if we had acted earlier, humankind wouldn't have found themselves in this predicament.*





Learning Points

The Passive: The Present Continuous and Present Perfect Tenses

You use the Present Continuous to talk about an action that is happening at the time of speaking. The action or event is in progress.

*As a result, the atmosphere **is trapping** more heat, and the earth **is warming up**.*

You use the Present Perfect to talk about a finished action or event with present consequences. There is a connection between what happened in the past and the present.
*Many scientists say that human activities **have caused** greenhouse gases to build up in the atmosphere.*

You often choose the passive form when you want to talk about an action but are not so interested in saying who or what does/did it. Passives without *agents* (the doers) are common in academic and scientific writing.

Look at the chart below for the structure of the passive voice.

Name	Structure	Example
Present Continuous	<i>am/are/is + being + past participle</i>	<i>Whenever we burn fossil fuels, more greenhouse gases are being released into the atmosphere.</i>
Present Perfect	<i>have/has + been + past participle</i>	<i>Once greenhouse gases have been released into the atmosphere, most of them remain there for decades and continue to trap heat.</i>

Examples:

- *This has been called the "greenhouse effect", which makes it possible for humans, animals, and plants to live.*
- *Once greenhouse gases have been released into the atmosphere, most of them remain there for decades and continue to trap heat.*
- *According to this theory, whenever we burn fossil fuels, such as gasoline, oil, coal, or natural gas, more greenhouse gases are being released into the atmosphere.*
- *For this reason, plants are being grown in glass greenhouses.*
- *As a result, the atmosphere is trapping more heat, and the earth is warming up.*

